Supplementary Materials

Cost-Effectiveness of Pembrolizumab in Combination with Chemotherapy as Neoadjuvant Treatment and Continued as a Single Agent Adjuvant Treatment for High-risk Early-stage Triple-negative Breast Cancer in Hong Kong

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Table S1 Scenario analysis results

Scenarios	**ICER (ΔCost/ΔQALY)**
EFS	
EFS function – Pembrolizumab + chemotherapy- Piecewise - Week 50 - Log-normal	313,297
EFS function - Pembrolizumab + chemotherapy - Piecewise - Week 68 - Log-normal	181,014
EFS function - Pembrolizumab + chemotherapy - Piecewise - Week 43 - Log-normal	239,261
EFS function - Chemotherapy - Piecewise - Week 50 - Generalized Gamma	148,372
EFS function - Chemotherapy - Piecewise - Week 68 - Log-normal	150,208
EFS function - Chemotherapy - Piecewise - Week 43 - Log-normal	93,971
Utility	
Utility algorithm (UK 5L crosswalk to 3L)	145,344
TOT	
TOT measure - Pembrolizumab + chemotherapy - KM lower 95% CI	128,833
TOT measure - Pembrolizumab + chemotherapy - KM upper 95% CI	142,138
TOT measure - Chemotherapy - KM lower 95% CI	135,469
TOT measure - Chemotherapy - KM upper 95% CI	134,940
Costs	
DM treatment patterns based on KEYNOTE-355	114,459
Time horizon (10 years)	664,935
No half-cycle correction	134,394
Allow vial-sharing	135,345
Do not apply relative dose intensity	140,359
Annual discount rate - costs (0%)	118,020
Annual discount rate - costs (5%)	143,379
Annual discount rate - effectiveness (0%)	81,902
Annual discount rate - effectiveness (5%)	182,269
Remission after 8 years	204,356

No AE costs	134,881
Include indirect costs	136,994

Abbreviation: 3L, Three-level; 5L, Five-level; AE, Adverse Event; CI, Confidence Interval; DM, Disease Management; EFS, Event-free Survival; ICER, Incremental Cost-Effectiveness Ratio; KM, Kaplan-Meier; pembrolizumab + chemotherapy, pembrolizumab in combination with chemotherapy as neoadjuvant treatment followed by adjuvant pembrolizumab; QALY, Quality-Adjusted Life Years; TOT, Time on Treatment; UK, United Kingdom.

Table S2 Probabilistic sensitivity analysis and one-way sensitivity analysis parameters

Parameter	**Base-case**	**DSA range**	**PSA Distribution**	**Notes for SE**
Female weight (kg) - mean	76.37	95% CI	Normal	The standard deviation for weight was calculated by multiplying the Hong Kong specific mean weight with the ratio of SD of weight and mean weight sourced from KN522 trial
Body surface area (m2) - mean	1.92	95% CI	Normal	The standard deviation for BSA was calculated by multiplying the Hong Kong specific mean BSA with the ratio of SD of BSA and mean BSA sourced from KN522 trial
EFS - Pembrolizumab + chemotherapy - Piecewise - 50 - Generalized Gamma - A	5.28	95% CI	Multivariate normal	Not applicable; SA inputs based on Cholesky decomposition matrix
EFS - Pembrolizumab + chemotherapy - Piecewise - 50 - Generalized Gamma - B	1.37	95% CI	Multivariate normal	Not applicable; SA inputs based on Cholesky decomposition matrix
EFS - Pembrolizumab + chemotherapy - Piecewise - 50 - Generalized Gamma - C	-4.27	95% CI	Multivariate normal	Not applicable; SA inputs based on Cholesky decomposition matrix

EFS - Chemotherapy - Piecewise - 50 - Log-normal - A	7.15	95% CI	Multivariate normal	Not applicable; SA inputs based on Cholesky decomposition matrix
EFS - Chemotherapy - Piecewise - 50 - Log-normal - B	0.9945	95% CI	Multivariate normal	Not applicable; SA inputs based on Cholesky decomposition matrix
Exponential rate of LR to DM or death	0.0133	95% CI	Normal	-
% transition from LR to DM	0.9000	95% CI	Beta	α and β were obtained from KEYNOTE-522 trial data.
Exponential rate of DM to death - Pembrolizumab + chemotherapy	0.0192	95% CI	Normal	SE assumed to be equal to 50% of the base-case value
Exponential rate of DM to death - Chemotherapy	0.0166	95% CI	Normal	SE assumed to be equal to 50% of the base-case value
% LR among first EFS event (Year 1) - Pembrolizumab + chemotherapy	0.3654	95% CI	Beta	-
% LR among first EFS event (Year 1) - Chemotherapy	0.4483	95% CI	Beta	-
% DM among first EFS event (Year 1) - Pembrolizumab + chemotherapy	0.4808	95% CI	Beta	-
% DM among first EFS event (Year 1) - Chemotherapy	0.5172	95% CI	Beta	-

% LR among first EFS event (Year 2+) - Pembrolizumab + chemotherapy	0.2676	95% CI	Beta	-
% LR among first EFS event (Year 2+) - Chemotherapy	0.2813	95% CI	Beta	-
% DM among first EFS event (Year 2+) - Pembrolizumab + chemotherapy	0.6338	95% CI	Beta	-
% DM among first EFS event (Year 2+) - Chemotherapy	0.6406	95% CI	Beta	-
% received initial surgery - Pembrolizumab + chemotherapy	0.9800	95% CI	Beta	-
% received initial surgery - Chemotherapy	0.9770	95% CI	Beta	-
% received radiation - Pembrolizumab + chemotherapy	0.7590	95% CI	Beta	-
% received radiation - Chemotherapy	0.7850	95% CI	Beta	-
EF on treatment utility - Pembrolizumab + chemotherapy	0.8590	95% CI	Beta	-
EF on treatment utility - Chemotherapy	0.8590	95% CI	Beta	-

EF off treatment utility - Pembrolizumab + chemotherapy	0.8620	95% CI	Beta	-
EF off treatment utility - Chemotherapy	0.8620	95% CI	Beta	-
LR utility - Pembrolizumab + chemotherapy	0.8000	95% CI	Beta	-
LR utility - Chemotherapy	0.8000	95% CI	Beta	-
DM utility - Pembrolizumab + chemotherapy	0.6970	95% CI	Beta	-
DM utility - Chemotherapy	0.6970	95% CI	Beta	-
Grade 3+ AE utility decrement - Pembrolizumab + chemotherapy	0.0250	95% CI	Beta	-
Grade 3+ AE utility decrement - Chemotherapy	0.0250	95% CI	Beta	-
Age-adjusted disutility coefficient- Age (years)	-0.0003	95% CI	Normal	SE assumed to be equal to 50% of the base-case value
Age-adjusted disutility coefficient - Age2	-0.00003	95% CI	Normal	SE assumed to be equal to 50% of the base-case value
Age-adjusted disutility coefficient - Male	0.0212	95% CI	Normal	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Carboplatin	0.4467	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Paclitaxel	6.60	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value

Drug unit cost - Doxorubicin	1.56	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Epirubicin	2.72	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Cyclophosphamide	0.1090	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Capecitabine	0.0070	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Gemcitabine	0.1290	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Nab-paclitaxel	47.95	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Atezolizumab	16.27	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Drug unit cost - Bevacizumab	45.20	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Admin unit cost - Chemotherapy preparation and administration (per day)	2,400	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Total metastatic treatment costs - Pembrolizumab + chemotherapy	308,031	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Total metastatic treatment costs - Chemotherapy	206,854	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Surgery cost	29,769	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value

Radiation cost	150,000	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Disease management costs in EF state (per week, year 1-3)	50.58	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Disease management costs in EF state (per week, year 4-5)	50.58	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Disease management costs in EF state (per week, year 6-10)	25.29	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Disease management costs in EF state (per week, year 11+)	10.12	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Disease management costs in LR state entry (one-off)	9,410	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Disease management costs in LR state subsequent week (per week)	2,834	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Disease management costs in DM state entry (one-off)	9,410	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Disease management costs in DM state subsequent week (per week)	7,163	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Terminal care costs from EF or LR to death (one-off)	261,964	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
Terminal care costs from DM to death (one-off)	261,964	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value
AE costs - Pembrolizumab + chemotherapy	5,043	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value

AE costs - Chemotherapy	4,261	95% CI	Gamma	SE assumed to be equal to 50% of the base-case value

Abbreviation: AE, Adverse Event; CI, Confidence Interval; DM, Distant Metastases; EF, Event-free; EFS, Event-free Survival; LR, Locoregional Recurrence; PSA, Probabilistic Sensitivity Analysis; SA, Sensitivity Analysis; SD, Standard Deviation; SE, Standard Error.

NOTE: The drug costs provided in table above are unit cost per mg. Also, costs are mentioned in Hong Kong Dollar (HKD).

Table S3 Summary of health state transition options provided in the economic model

Transition(s)	**Estimation approach**	**Data source(s)**	**Scenario or one-way sensitivity analyses performed**
EF → LR EF → DM EF → Death	Time dependent transition probabilities were estimated based on 1) extrapolated EFS and 2) proportion of LR, DM and death as the first EFS event ▪ Using the patient-level data from KEYNOTE-522, EFS was extrapolated based on parametric functions for each arm. ▪ No remission assumption was applied in the base-case. ▪ No treatment waning effect was considered in the base-case. ▪ Probability of experiencing LR, DM or death per cycle were estimated from EFS. ▪ The probability of patients experiencing LR, DM and death as the first EFS event in Year 1 and Year 2+, respectively, were obtained from the KEYNOTE-522 clinical trial. ▪ The transition probabilities of EF → LR, EF → DM and EF → death were then	▪ KEYNOTE-522 ▪ Life tables for Hong Kong (2022) * - for transitions to death	▪ Alternative parametric distributions

	calculated based on the probability of experiencing event (LR, DM or death) and the proportions of each event. ▪ The transition probabilities of EF → death were constrained to be at least as high as the all-cause natural mortality for female population.		
LR → DM LR → Death	Transition probabilities starting from LR were assumed to be equivalent between arms, and constant across all cycles. ▪ The transition probabilities of LR → DM or death were obtained from the KEYNOTE-522 clinical trial by pooling data from the two treatment arms. ▪ The proportions of patients experiencing DM and death, were obtained from the KEYNOTE-522 clinical trial. ▪ The transition probabilities of LR → DM, and LR → death was calculated based on the probability of experiencing either event (DM or death) and the proportions of each event. ▪ The transition probabilities of LR → death was constrained to be at least as high as the all-cause natural mortality for female population.	▪ KEYNOTE-522 ▪ Life tables for Hong Kong (2022) * - for transitions to death	▪ Not applicable (NA)
DM → Death	Transition probability from DM → death was estimated based on the treatment rate, the	▪ KEYNOTE-522	▪ Using KEYNOTE-355 as the source in the DM

	expected mix of first line (1L) treatments in the DM state, and the efficacy of these 1L treatments in terms of mean OS ▪ KEYNOTE-522 was selected as the base-case source to estimate mean OS of all patients following distant metastases. Therefore, no separate treatment rate and expected treatment mix needed to be considered. ▪ The transition probabilities of DM → death were constrained to be at least as high as the all-cause natural mortality for female population.	▪ Life tables for Hong Kong (2022) * - for transitions to death	state, the transition probabilities of DM → death was derived based on assumptions and inputs related to 1) rechallenge with pembrolizumab or other immune-oncology (I-O)-agent; 2) PD-L1 testing and positive rate; 3) treatment rate; 4) treatment mix for PD-L1 positives and PD-L1 negatives in the metastatic setting; and 5) mean OS of patients who received each 1L treatment and who did not receive the 1L treatments.

Abbreviations: 1L. 1 Line; DM, Distant Metastases; EF, Event-free; EFS, Event-free Survival; I-O, Immuno-Oncology; LR, Locoregional Recurrence; NA, Not Available; OS, Overall Survival; PD-L1, Programmed Death-Ligand 1.

*Source for Hong Kong Life Table: Census and Statistics Department, The Government of the Hong Kong Special Administrative Region, 22 December 2022

Table S4 Age-related dis-utility

Parameter	**Coefficient**
Age (years)	-0.0002587
Age^2	-0.0000332

Male	0.0212126
Intercept	0.9508566

Source: Ara R, Brazier JE. Populating an economic model with health state utility values: moving toward better practice. Value in Health. 2010

Table S5 All-cause grades 3+ AEs with incidence ≥ 5%

Grade 3+ AE	**Pembrolizumab + chemotherapy**	**Chemotherapy**	**Average duration of AEs (weeks)**	**Source**
Neutropenia	35.20%	34.40%	12.53	KEYNOTE-522 (Cut-off date: 23 March 2021)
Neutrophil count decreased	19.00%	23.70%	12.53	
Anemia	19.50%	15.70%	12.53	
Febrile neutropenia	18.40%	16.20%	12.53	
White blood cell count decreased	7.80%	5.40%	12.53	
Alanine aminotransferase increased	6.40%	2.80%	12.53	

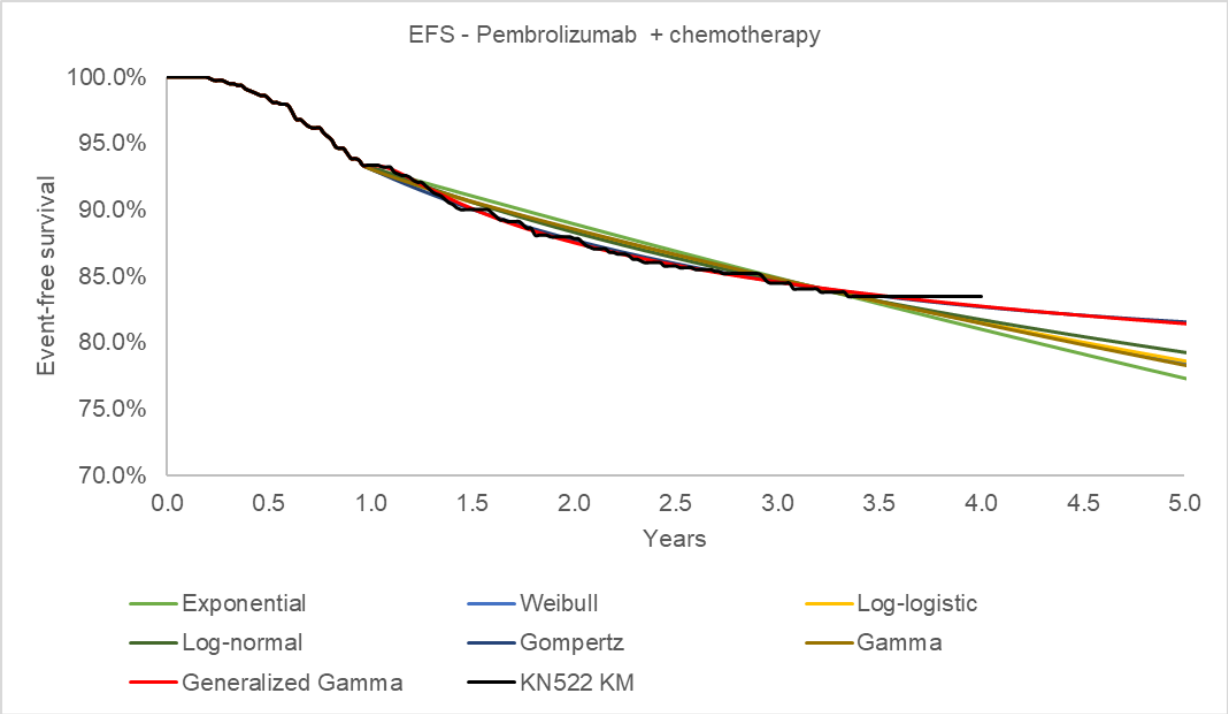
Abbreviation: AEs, Adverse Events.

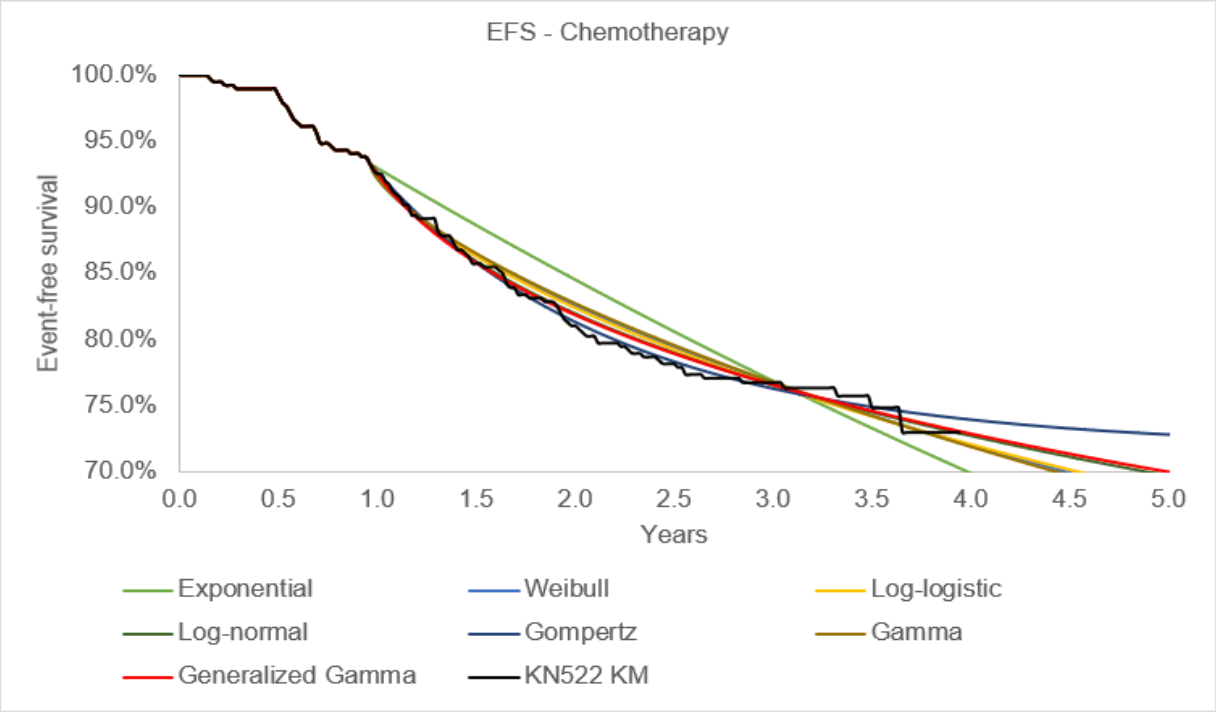
Table S6 Percent of patient hospitalized per AE

Grade 3+ AE	**Pembrolizumab + chemotherapy**	**Chemotherapy**	**Source**
Neutropenia	3.30%	0.80%	KEYNOTE-522 (Cut-off date: 23 March 2021)
Neutrophil count decreased	1.30%	0.00%	
Anemia	12.00%	13.10%	
Febrile neutropenia	80.40%	74.60%	
White blood cell count decreased	0.00%	9.50%	
Alanine aminotransferase increased	6.10%	0.00%	

Abbreviation: AE, Adverse Event.

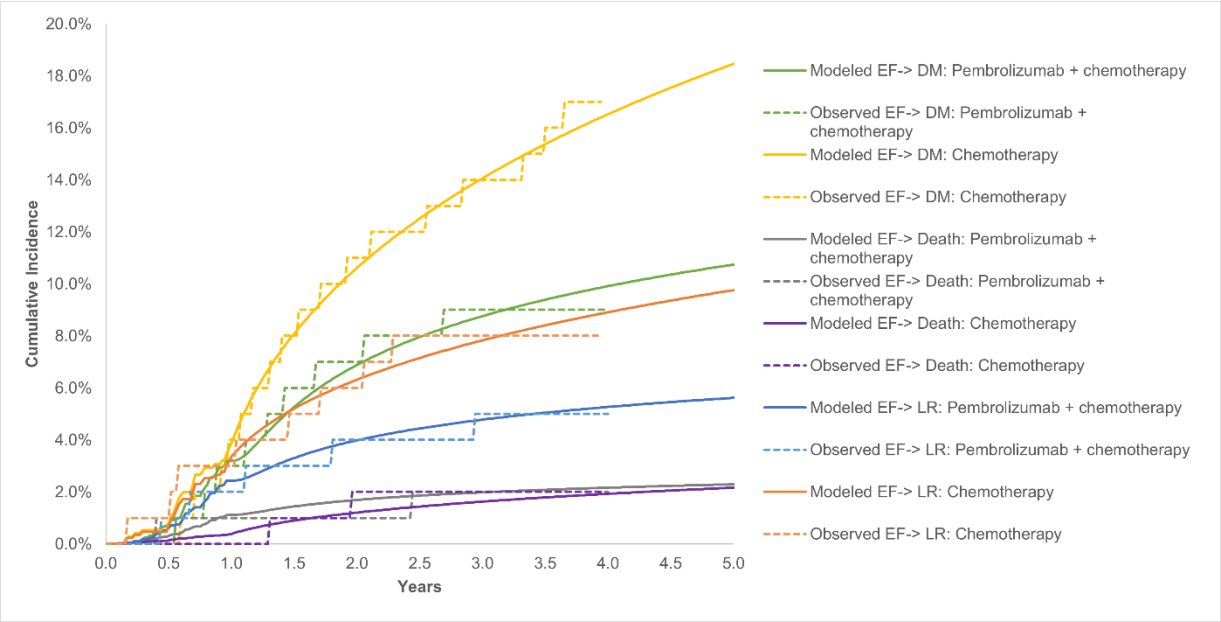
Fig. S1 Events-free survival base-case curves and extrapolations using alternative parametric functions





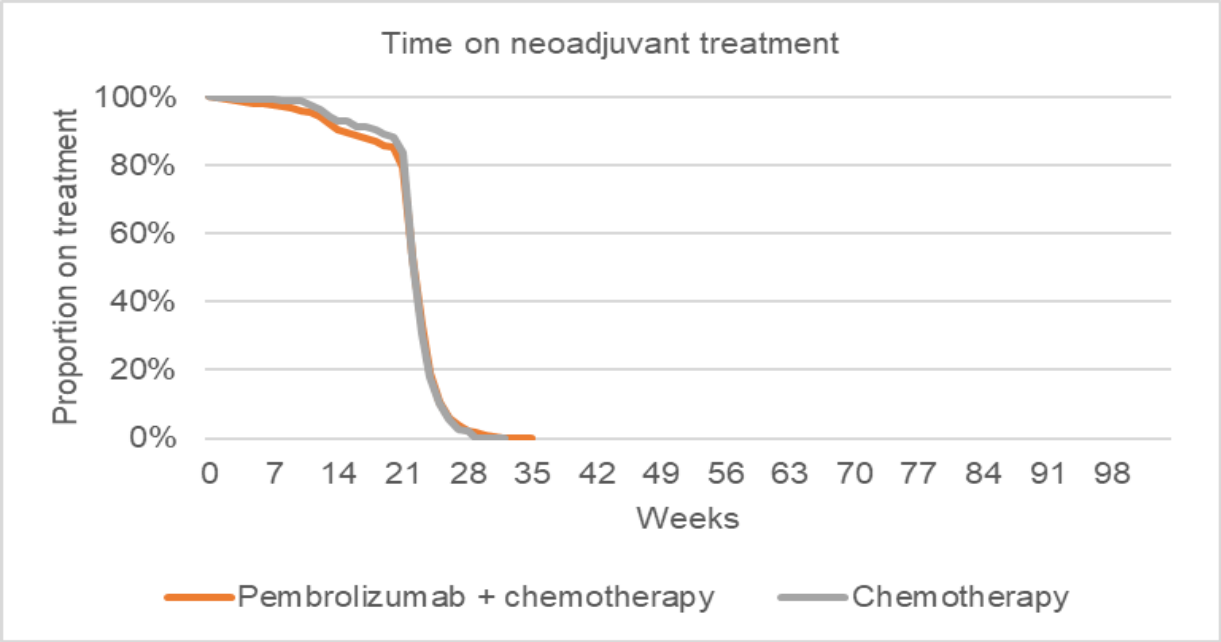
Abbreviation: EFS, Event-free Survival; KN522, KEYNOTE-522

Fig. S2 Cumulative incidence vs. observed incidence of EF → LR, EF → DM and EF → death for pembrolizumab + chemotherapy



Abbreviations: DM, Distant Metastasis; EF, Event-free; LR, Locoregional Recurrence

Fig. S3 Time on neoadjuvant and adjuvant treatment setting



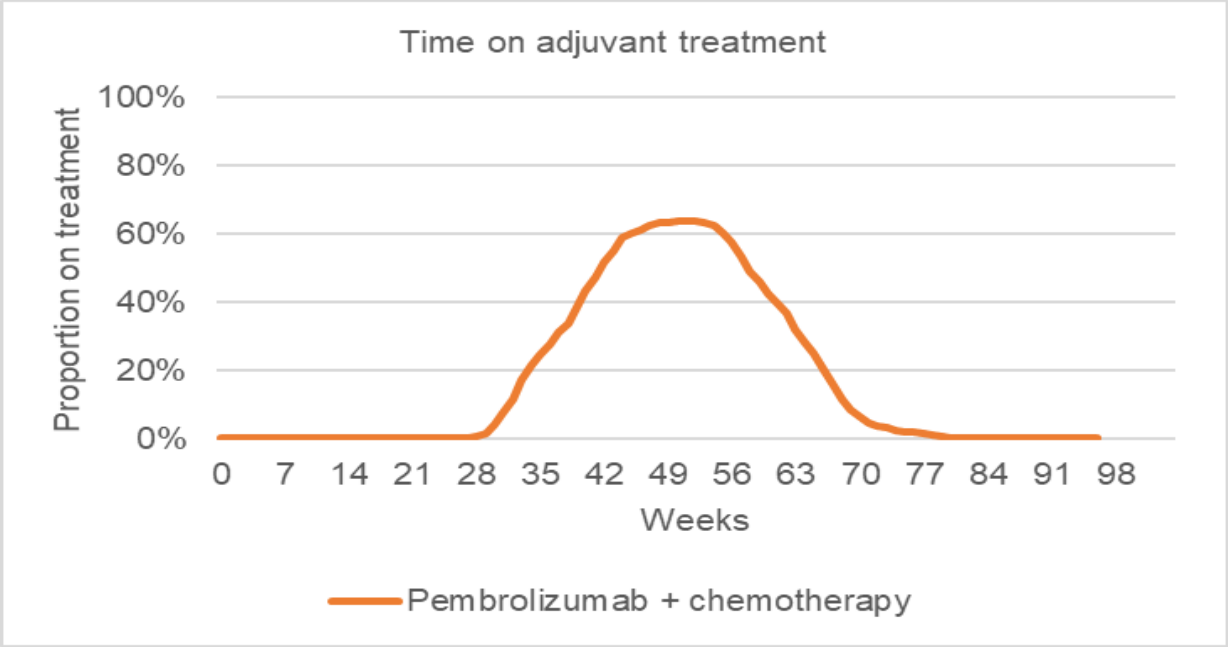
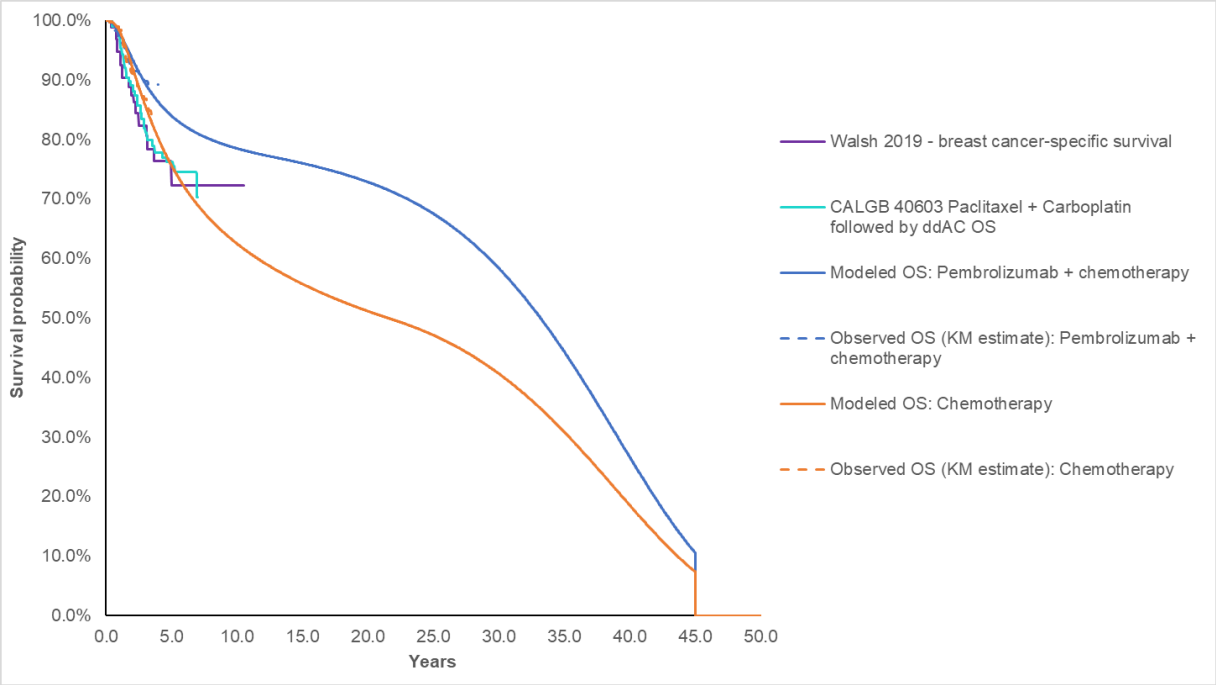


Fig. S4 External validation of modeled OS



Abbreviation: ddAC: Dose-Dense Doxorubicin and Cyclophosphamide; KM, Kaplan Merrie, OS, Overall Survival